

PES UNIVERSITY

Department of CSE

Software Engineering Laboratory – Lab 1
Requirements Engineering & UML Use-Case Modelling

Problem Statement #04

Campus & Academic Operations

Student Project Portfolio & Showcase App

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1. Requirements Table

Student Project Portfolio & Showcase App

1.1 Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable student teams to create and publish project showcase profiles containing an abstract, demo video link, and verified GitHub repository URL.	High	Pass: Profile is published when all required information and a valid repository URL are provided. Fail: Profile is published with an invalid repository URL.	Allows student teams to present their projects in a standardized format.
FR-002	Functional	The system shall enable student teams to upload and manage multimedia project artifacts such as images, presentations, and demonstration videos.	High	Pass: Valid multimedia files are successfully uploaded and associated with the project. Fail: Unsupported or invalid files are accepted.	Provides judges and visitors with sufficient project information for evaluation.
FR-003	Functional	The system shall enable panel judges to evaluate submitted projects using predefined digital grading rubrics.	High	Pass: Judge can assign scores according to each rubric criterion and submit the evaluation. Fail: Evaluation is submitted without required rubric scores.	Ensures consistent and structured project evaluation.
FR-004	Functional	The system shall enable verified public users to vote for eligible projects while preventing multiple votes from the same user for the same project.	High	Pass: A verified user can cast one vote per eligible project. Fail: The same user can cast multiple votes for the same project.	Provides fair public participation while reducing fraudulent voting.
FR-005	Functional	The system shall calculate and display project rankings and leaderboards based on judge scores and	Medium	Pass: Leaderboard reflects updated scores and votes correctly. Fail: Leaderboard displays	Allows participants and visitors to view project performance and standings.

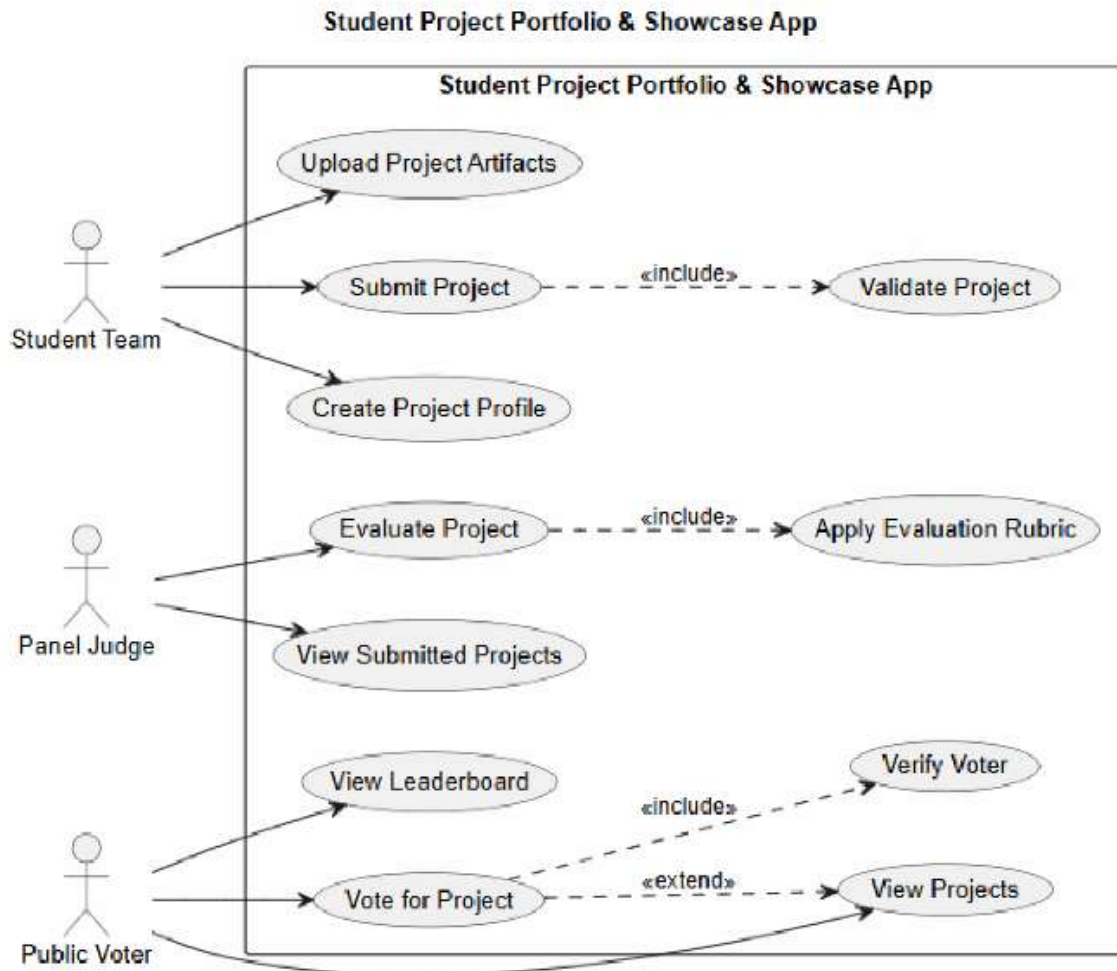
		public voting results.		outdated or incorrectly calculated rankings.	
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1.2 Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The public voting and leaderboard calculations shall update in real-time with an API response latency below 300 ms under normal and peak expected load.	High	Pass: Benchmarking confirms API response latency remains below 300 ms under simulated peak load. Fail: Response latency exceeds 300 ms.	Ensures responsive voting and leaderboard functionality.
NFR-002	Security	The system shall implement authentication and anti-sybil mechanisms to ensure that only verified users can vote and that fraudulent or duplicate voting attempts are prevented.	High	Pass: Duplicate and unverified voting attempts are rejected during security testing. Fail: An unverified user or duplicate account can successfully cast additional votes.	Protects the fairness and integrity of the public voting system.

2. UML Use-Case Diagram

Insert the completed UML Use-Case Diagram below. The diagram must model all actors, primary use cases, and include at least one «include» and one «extend» relationship.



Actors

- Student Team
- Panel Judge
- Public Voter

Primary Use Cases

- Create Project Profile
- Upload Project Artifacts

- [Submit Project](#)
- [Validate Project](#)
- [View Submitted Projects](#)
- [Evaluate Project](#)
- [Apply Evaluation Rubric](#)
- [View Projects](#)
- [Vote for Project](#)
- [Verify Voter](#)
- [View Leaderboard](#)

3. Use-Case Flow Specification

UC-01: Submit Project

Primary Actor: Student Team

3.1 Preconditions

1. The student team is registered and authenticated.
2. The student team has created a project profile.
3. Required project information has been entered.
4. The project contains the required multimedia artifacts and repository information.

3.2 Postconditions

Success:

- The project is successfully submitted.
- The project is validated.
- The project becomes available in the showcase catalog for eligible judges and public users.

Failure:

- The project remains unpublished.
- The student team is informed about the validation error.

3.3 Main Success Scenario

5. Student Team logs into the showcase application.
6. Student Team selects Create/Manage Project.
7. Student Team enters the project title and abstract.
8. Student Team adds the demo video link.
9. Student Team provides the GitHub repository URL.
10. Student Team uploads the required project artifacts.
11. Student Team selects Submit Project.
12. System validates the project information.
13. System verifies that the GitHub repository URL is valid.
14. System confirms successful validation.
15. System publishes the project to the showcase catalog.
16. System displays a confirmation message to the Student Team.

3.4 Alternate Flow

A1 — Invalid GitHub Repository URL

17. During validation, the system detects that the GitHub repository URL is invalid or inaccessible.
18. The system rejects the submission.
19. The system displays an error message requesting a valid repository URL.
20. Student Team corrects the URL.
21. Student Team resubmits the project.
22. System performs validation again.